BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Mathematics (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** 7191

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The quadratic formula is:

(A) $x = (-b \pm \sqrt{(b^2 - 4ac)})/2a$

(B) $x = (b \pm \sqrt{(b^2 - 4ac)})/2a$

(C) $x = (-b \pm \sqrt{(b^2 + 4ac)})/2a$

(D) $x = (b \pm \sqrt{(b^2 + 4ac)})/2a$

درج ذیل فارمولا ہے:

Q2. If α, β are roots of equation $7x^2 - x + 4 = 0$ then αβ is:

(A) –1/7

(B) 4/7

(C) 7/4

(D) –4/7

اگر α, β مساوات $7x^2 - x + 4 = 0$ کے روٹس ہوں تو $\alpha\beta$ برابر ہے:

Q3. Sum of cube roots of unity is:

(A) 0

(B) 1

(C) –1

(D) 3

اکائی کے جذر المکعب کا مجموعہ ہے:

Q4. In a ratio a : b, a is called:

(A) Relation
(B) Antecedent
(C) Consequent
(D) Ratio

نسبت a : b میں a کہلاتا ہے:

[illegible in source scan] **(A)**
[illegible in source scan] **(B)**
[illegible in source scan] **(C)**
نسبت **(D)**

Q5. Find x in proportion 4 : x :: 5 : 15 :

(A) 75/4
(B) 4/3
(C) 3/4
(D) 12

تناسب 4 : x :: 5 : 15 میں x معلوم کیجیے:

Q6. Partial fractions of $(x^2 + 1)/((x + 1)(x - 1))$ are of the form:

(A) 1 + A/(x+1) + (Bx+C)/(x–1)
(B) A/(x+1) + B/(x–1)
(C) (Ax+B)/(x+1) + C/(x–1)
(D) 1 + A/(x+1) + B/(x–1)

$(x^2+1)/((x+1)(x-1))$ کی جزوی کسریں اس شکل کی ہوتی ہیں:

Q7. A collection of well defined objects is called:

(A) Sub set
(B) Power set
(C) Set
(D) Empty set

اچھی طرح سے متعین اشیاء کا مجموعہ کہلاتا ہے:

سب سیٹ **(A)**
پاور سیٹ **(B)**
سیٹ **(C)**
خالی سیٹ **(D)**

Q8. If $A \subseteq B$ then $A \cap B$ is equal to:

(A) A
(B) B
(C) φ
(D) U

اگر $A \subseteq B$ ہو تو $A \cap B$ برابر ہوتا ہے:

Q9. A grouped frequency table is called:

(A) Data
(B) Frequency distribution
(C) Frequency polygon
(D) Histogram

گروپڈ فریکوئنسی ٹیبل کہلاتا ہے:

(A) ڈیٹا
(B) فریکوئنسی ڈسٹری بیوشن
(C) فریکوئنسی پولی گان
(D) ہسٹوگرام

Q10. Mean is affected by change in: _____

(A) Value
(B) Ratio
(C) Origin
(D) Proportion

حسابی اوسط _____ تبدیل کرنے سے متاثر ہوتا ہے۔

(A) قیمت
(B) نسبت
(C) مبدا
(D) تناسب

Q11. $\sec^2 \theta =$ _____

(A) $1 - \sin^2 \theta$
(B) $1 + \tan^2 \theta$
(C) $1 + \cos^2 \theta$
(D) $1 - \tan^2 \theta$

Q12. Radii of a circle are:

(A) All equal
(B) Double of the diameter
(C) All unequal
(D) Half of any chord

ایک ہی دائرے کے رداس ہوتے ہیں:

(A) تمام برابر
(B) قطر سے دوگنا
(C) تمام غیر برابر
(D) کسی بھی وتر سے آدھے

Q13. A circle has only one: _____

(A) Secant
(B) Chord
(C) Diameter
(D) Center

ایک دائرے کا صرف ایک ہی _____ ہوتا ہے۔

(A) قاطع
(B) وتر
(C) قطر
(D) مرکز

Q14. A 4cm long chord subtends a central angle of 60° the radial segment of this circle is: _____

(A) 1 (B) 2

(C) 3 (D) 4

ایک 4cm لمبی وتر مرکزی زاویہ 60° کا زاویہ بناتا ہے اس دائرے کا رداسی قطعہ _____ ہوگا۔

Q15. The circumference of circle is called:

(A) Chord (B) Segment

(C) Boundary (D) Radius

دائرے کے محیط کو کہا جاتا ہے:

(A) وتر **(B)** قطعہ

(C) حد **(D)** رداس

Mathematics (Subjective) Group - I

Time Allowed: 02:10 Hours | **Maximum Marks:** 60 | **Paper Code:** 7191

Note: Part - I (حصہ اول)

Q2. (Marks 12) Write short answers to any SIX parts.

(i) Write the equation in standard form: $(x+4)/(x-2) - (x-2)/x + 4 = 0$

(ii) Solve the equation: $x^2 + 2x - 2 = 0$

(iii) Solve: $\sqrt{(3x + 18)} = x$

(iv) Evaluate: $(1 - 3\omega - 3\omega^2)^5$

(v) Without solving find the sum and product of the roots of quadratic equation: $(\ell + m)x^2 + (m + n)x + n - \ell = 0$

(vi) Find ω^2 if: $\omega = (-1 + \sqrt{-3})/2$

(vii) If $y \propto 1/x$ and $y = 4$ when $x = 3$ find x when $y = 24$

(viii) Find the third proportional of: $(p^2 - q^2)/(p^3 + q^3)$, $(p - q)/(p^2 - pq + q^2)$

(ix) Find the fourth proportional of: $4x^4$, $2x^3$, $18x^5$

Q3. (Marks 12) Write short answers to any SIX parts.

(i) What is proper fraction?

(ii) How can we make partial fractions of $(x - 2)/((x + 2)(x + 3))$?

(iii) If X = {1, 4, 7, 9}, Y = {2, 4, 5, 9} then find Y ∪ X

(iv) Define a function.

(v) If X = {1, 3, 5, 7, ..., 19}, Y = {0, 2, 4, 6, ..., 20} and Z = {2, 3, 5, 7, 11, 13, 17, 19, 23} then find X ∩ (Y ∪ Z).

(vi) Write all the subsets of {a, b}.

(vii) Find arithmetic mean: 200, 225, 350, 375, 270, 320, 290

(viii) Define a frequency distribution.

(ix) Define mode.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Define ratio and give one example.

(ii) Find a third proportional to 28 and 4.

(iii) Define an angle.

(iv) Convert $-7\pi/8$ to degrees.

(v) Find r when $\ell = 56$cm and $\theta = 45°$

Note: (بقیہ اگلے صفحے پر)

(vi) Prove that: $(1 - \sin^2\theta)(1 + \tan^2\theta) = 1$

(vii) Express $-67.58°$ into D°, M' and S'' form.

(viii) In a ΔABC, calculate mBC when mAB = 5cm, mAC = 4cm and $m\angle A = 60°$

(ix) Divide an arc of any length into two equal parts.

Note: Part - II, Attempt any THREE questions. Each question carries 08 marks. Question No. 9 is compulsory.

حصہ دوم، کوئی سے تین سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 08 نمبر ہیں۔ سوال نمبر 9 لازمی ہے۔

Q5(a). (Marks 04) Solve the equation: $2x + 5 = \sqrt{(7x + 16)}$

Q5(b). (Marks 04) If α, β are the roots of the equation $x^2 - 3x + 6 = 0$, form equation whose roots are α^2, β^2.

Q6(a). (Marks 04) Using theorem of componendo dividendo, find the value of $(x - 3y)/(x + 3y) : (x + 3z)/(x - 3z)$ if $x = 3yz/(y - z)$

Q6(b). (Marks 04) Resolve into partial fractions: $(x^2 - 3x + 1)/((x - 1)^2(x - 2))$

Q7(a). (Marks 04) If $L = \{x \mid x \in N \wedge x \leq 5\}$, $M = \{y \mid y \in P \wedge y < 10\}$, then make the relation from L to M $R = \{(x, y) \mid y = x\}$

Q7(b). (Marks 04) Find the standard deviation of the data: 9, 3, 8, 8, 9, 8, 9, 18

Q8(a). (Marks 04) Verify: $(\tan \theta + \cot \theta)(\cos \theta + \sin \theta) = \sec \theta + \operatorname{cosec} \theta$

Q8(b). (Marks 04) Draw circle which touches both the arms of angle 45°.

Q9. (Marks 08) Prove that if two chords of a circle are congruent then they will be equidistant from the center. — OR — Prove that two tangents drawn to a circle from a point outside it, are equal in length.

GROUP – II

Mathematics (Objective) Group - II

Time Allowed: 20 Minutes	**Maximum Marks:** 15	**Paper Code:** 7198

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. A portion of a circle between two radii and an arc is called:

(A) Sector
(B) Segment
(C) Chord
(D) Boundary

دو رداسوں اور قوس کے درمیان دائرے کا حصہ کہلاتا ہے:

(A) سیکٹر
(B) قطعہ
(C) وتر
(D) حد

Q2. An arc subtends a central angle of 40°, then corresponding chord will subtend a central angle of:

(A) 20°
(B) 40°
(C) 60°
(D) 80°

ایک قوس مرکزی زاویہ 40° بناتی ہے تو اس کے متعلقہ وتر مرکزی زاویہ بنائے گا:

Q3. A circle has only one:

(A) Secant
(B) Chord
(C) Diameter
(D) Center

ایک دائرے کا صرف ایک ہی ہوتا ہے:

(A) قاطع
(B) وتر
(C) قطر
(D) مرکز

Q4. A chord passing through the center of a circle is called:

(A) Radius
(B) Circumference
(C) Line segment
(D) Diameter

دائرے کے مرکز سے گزرنے والا وتر کہلاتا ہے:

(A) رداس
(B) محیط
(C) خطی قطعہ
(D) قطر

Q5. $\sec^2 \theta =$ _____

(A) $1 - \sin^2 \theta$
(B) $1 + \tan^2 \theta$
(C) $1 + \cos^2 \theta$
(D) $1 - \tan^2 \theta$

Q6. A data in the form of a frequency distribution is called:

(A) Grouped data
(B) Ungrouped data
(C) Histogram
(D) Ratio

تعدد کی شکل میں ڈیٹا کہلاتا ہے:

(A) گروپڈ ڈیٹا
(B) ان گروپڈ ڈیٹا
(C) ہسٹوگرام
(D) نسبت

Q7. A measure which determines the middle most observation in a data set is called:

(A) Mean
(B) Mode
(C) Harmonic mean
(D) Median

ایسا پیمانہ جو ڈیٹا سیٹ میں درمیانی مشاہدے کا تعین کرتا ہے کہلاتا ہے:

(A) [illegible in source scan]
(B) موڈ
(C) ہارمونک اوسط
(D) میڈین

Q8. Power set of an empty set is:

(A) {φ}
(B) {a}
(C) {φ, a}
(D) {φ, {φ}}

خالی سیٹ کا پاور سیٹ ہے:

Q9. Point (–1, 4) lies in the quadrant:

(A) I
(B) II
(C) III
(D) IV

نقطہ $(-1, 4)$ ربع میں ہوتا ہے:

Q10. $(x + 3)^2 = x^2 + 6x + 9$ is: _____

(A) A linear equation
(B) An equation
(C) An identity
(D) Fraction

$(x+3)^2 = x^2 + 6x + 9$ ایک _____ ہے۔

(A) ایک خطی مساوات
(B) ایک مساوات
(C) ایک شناخت
(D) کسر

Q11. Find the value of x in proportion 4 : x :: 5 : 15

(A) 75/4
(B) 4/3
(C) 3/4
(D) 12

تناسب $4 : x :: 5 : 15$ میں x کی قیمت معلوم کیجیے:

Q12. In a ratio x : y, y is called:

(A) Relation
(B) Antecedent
(C) Consequent
(D) Means

نسبت $x : y$ میں y کہلاتا ہے:

(A) تعلق
(B) مقدم
(C) موخر
(D) وسطین

Q13. If α, β are the roots of $x^2 - x - 1 = 0$ then product of the roots of 2α and 2β is:

(A) –2
(B) 2
(C) 4
(D) –4

اگر α, β مساوات $x^2 - x - 1 = 0$ کے روٹس ہوں تو 2α اور 2β کے روٹس کا حاصل ضرب ہے:

Q14. If $b^2 - 4ac < 0$, then the roots of the equation $ax^2 + bx + c = 0$ are:

(A) Rational
(B) Irrational
(C) Real
(D) Imaginary

اگر $b^2 - 4ac < 0$ ہو تو مساوات $ax^2 + bx + c = 0$ کے روٹس ہوتے ہیں:

(A) ناطق
(B) غیر ناطق
(C) حقیقی
(D) تخیلی

Q15. The number of methods to solve a quadratic equation is:

(A) 1
(B) 2
(C) 3
(D) 4

دو درجی مساوات کو حل کرنے کے طریقے ہیں:

Mathematics (Subjective) Group - II

Time Allowed: 02:10 Hours	**Maximum Marks:** 60	**Paper Code:** 7198

Note: Part - I (حصہ اول)

Q2. (Marks 12) Write short answers to any SIX parts.

(i) Write the quadratic equation in standard form: $(x+4)/(x-2) - (x-2)/x + 4 = 0$

(ii) Solve by factorization: $5x^2 = 15x$

(iii) Define reciprocal equation.

(iv) Find the sum and the product of the roots of the quadratic equation: $x^3 - 5x + 3 = 0$

(v) Evaluate: $(1 - \omega + \omega^2)^6$

(vi) Discuss the nature of the roots of the equation: $x^2 + 6x - 1 = 0$

(vii) Find the value of p, if the ratios $2p + 5 : 3p + 4$ and $3 : 4$ are equal.

(viii) If $V \propto R^3$ and $V = 5$ when $R = 3$, find R when $V = 625$

(ix) Find the fourth proportional of: $4x^4$, $2x^3$, $18x^5$

Q3. (Marks 12) Write short answers to any SIX parts.

(i) Resolve into partial fractions: $(x - 11)/((x - 4)(x + 3))$

(ii) What is a proper fraction?

(iii) Define one-one function.

(iv) If A = N and B = W then find the value of A – B

(v) If Y = Z^+ and T = O^+, then find Y ∪ T

(vi) If M = {d, e, f, g}, then find two binary relations in M × M.

(vii) What is a histogram?

(viii) Find range of: 11500, 12400, 15000, 14500, 14800

(ix) Write three properties of arithmetic mean.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Define direct variation.

(ii) If u and v varies inversely and u = 8 when v = 3, find v when u = 12.

(iii) Convert 15° to radian.

(iv) Find r when ℓ = 56cm and θ = 45°.

(v) Define an angle.

Note: (بقیہ اگلے صفحے پر)

(vi) Locate the angle 22 1/2° in xy-plane.

(vii) Find r when ℓ = 4cm, θ = 1/4 radian.

(viii) In a ΔABC, calculate mBC when mAB = 5cm, mAC = 4cm and m∠A = 60°

(ix) Write the formula for finding the angle subtended by the side of a n-sided polygon at center of circle.

Note: Part - II, Attempt any THREE questions. Each question carries 08 marks. Question No. 9 is compulsory.

حصہ دوم، کوئی سے تین سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 08 نمبر کے ہیں۔ سوال نمبر 9 لازمی ہے۔

Q5(a). (Marks 04) Solve the equation by completing square: $4 - 8/(3x + 1) = (3x^2 + 5)/(3x + 1)$

Q5(b). (Marks 04) Prove: $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x + \omega y + \omega^2 z)(x + \omega^2 y + \omega z)$

Q6(a). (Marks 04) Find x, if 8 – x : 11 – x :: 16 – x : 25 – x

Q6(b). (Marks 04) Resolve into partial fractions: $(x^2 + 2x + 1)/((x - 2)(x + 3))$

Q7(a). (Marks 04) If X = {1, 3, 5, 7, ..., 19}, Y = {0, 2, 4, 6, 8, ..., 20} and Z = {2, 3, 5, 7, 11, 13, 17, 19, 23} then find (X ∪ Y) ∩ (X ∪ Z).

Q7(b). (Marks 04) Find the standard deviation 'S': 9, 3, 8, 8, 9, 8, 9, 18

Q8(a). (Marks 04) Verify the identity: $(\sin\theta + \cos\theta)/(\tan^2\theta - 1) = \cos^2\theta/(\sin\theta - \cos\theta)$

Q8(b). (Marks 04) Draw two common tangents to two touching circles of radii 2.5cm and 3.5cm.

Q9. (Marks 08) Prove that the measure of central angle of a minor arc of a circle, is double that of the angle subtended by the corresponding major arc. — OR — Prove that two tangents drawn to a circle from a point outside it, are equal in length.